

END-TO-END HEALTHCARE ANALYTICS PROJECT

HEALTHCARE ANALYTICS SYSTEM

Case Study: Axon Healthcare

Healthcare Data Analytics using

MySQL • Excel • Power BI • Tableau

Transforming Healthcare Data into Actionable Business Insights

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Project Overview

Executive Summary

The **Healthcare Analytics System** is an end-to-end data analytics project developed using the **Axon Healthcare dataset** to demonstrate practical implementation of database design, SQL analytics, and business intelligence. The project integrates patient, doctor, department, visit, treatment, laboratory, billing, and insurance data into a centralized relational database designed in **MySQL**.

A normalized database structure was implemented to ensure data consistency, integrity, and efficient data retrieval. Data cleaning and validation techniques were applied to identify duplicates, missing values, and inconsistent records before performing analytical queries.

More than **100 SQL queries** were developed using joins, aggregate functions, subqueries, Common Table Expressions (CTEs), window functions, views, stored procedures, indexes, and query optimization techniques to generate meaningful business insights.

The processed data was visualized through interactive dashboards developed in **Microsoft Excel, Power BI, and Tableau**, enabling analysis of hospital operations, patient demographics, doctor performance, financial performance, laboratory diagnostics, and insurance claims.

Overall, this project demonstrates how healthcare data can be transformed into actionable insights that support informed decision-making, operational efficiency, and strategic planning.

Project Objectives

The Healthcare Analytics System was developed with the following objectives:

- Design and implement a normalized relational healthcare database using MySQL.
- Store and manage patient, doctor, visit, treatment, laboratory, billing, and insurance data efficiently.
- Ensure data quality through data cleaning and validation techniques.
- Perform exploratory data analysis to understand healthcare trends and operational performance.
- Develop analytical SQL queries for operational, financial, and clinical reporting.
- Implement advanced SQL concepts such as CTEs, window functions, views, stored procedures, and indexes.
- Build interactive dashboards using Microsoft Excel, Power BI, and Tableau.
- Generate key performance indicators (KPIs) to monitor hospital performance.
- Provide meaningful business insights that support data-driven decision-making.

Data Source & Data Model

Data Source

This project utilizes the **Axon Healthcare Dataset**, which simulates the day-to-day operations of a healthcare organization. The dataset contains information related to patients, doctors, departments, hospital visits, treatments, laboratory tests, billing transactions, insurance policies, insurance providers, and insurance claims.

The data was imported into MySQL using the **Table Data Import Wizard**, where it was organized into a structured relational database for analysis and reporting.

Database Tables

Table Name	Records	Primary Key	Purpose
Patients	10,000	Patient_ID	Stores patient demographic information, personal details, contact information, blood group, medical history, and insurance details.
Doctors	1,000	Doctor_ID	Contains doctor profiles including specialization, department assignment, qualifications, years of experience, and contact information.

Department	5	Department_ID	Maintains hospital department details such as department name, specialty, and operational information.
Visit	10,000	Visit_ID	Records every patient visit including attending doctor, visit type, diagnosis, visit status, and appointment details.
Treatment	10,000	Treatment_ID	Stores treatment information including treatment type, procedure details, treatment duration, treatment cost, and associated visit.
Lab_Test	10,000	Lab_Test_ID	Maintains laboratory investigation details such as test name, test result, normal range, report status, and linked patient visit.
Billing	10,000	Bill_ID	Stores financial transactions including amount billed, patient payment, insurance coverage, outstanding balance, and payment status.
Insurance_Policy	10,000	Policy_Number	Contains insurance policy information including provider, coverage period, policy type, and linked patient details.
Insurance_Claims	10,000	Claim_ID	Tracks insurance claims including claim amount, approved amount, claim status, processing dates, and policy reference.
Insurance_Provider	10	Provider_ID	Stores information about insurance companies including provider name, contact details, and available insurance plans.

Data Model

The Healthcare Analytics database follows a **normalized relational data model** designed to ensure data consistency, integrity, and efficient data retrieval. The database consists of ten interconnected tables linked through **Primary Keys (PK)** and **Foreign Keys (FK)**, establishing one-to-one and one-to-many relationships between entities.

The data model integrates patient information with hospital visits, treatments, laboratory tests, billing records, insurance policies, insurance claims, healthcare providers, and departments. This structure enables seamless analytical reporting while minimizing redundancy and maintaining referential integrity.

Parent Entity	Child Entity	Relationship Type	Description
Department	Doctors	One-to-Many	A department can have multiple doctors.
Doctors	Visit	One-to-Many	A doctor can attend multiple patient visits.
Patients	Visit	One-to-Many	A patient can have multiple hospital visits.
Visit	Treatment	One-to-Many	A visit may include one or more treatments.
Visit	Lab_Test	One-to-Many	A visit may require one or more laboratory tests.
Visit	Billing	One-to-One	Each visit generates a corresponding billing record.
Insurance_Provider	Insurance_Policy	One-to-Many	One insurance provider offers multiple insurance policies.
Insurance_Policy	Insurance_Claims	One-to-Many	A policy may have multiple insurance claims over time.
Billing	Insurance_Claims	One-to-One	Each insurance claim is linked to a specific billing record.

Tools Used & Project Methodology

Tools & Technologies Used

The Healthcare Analytics System was developed using a combination of database management, data visualization, and business intelligence tools. Each tool was selected based on its capabilities to support different stages of the analytics workflow, from data storage and querying to dashboard development and business reporting.

Tool / Technology	Purpose	Key Features Utilized
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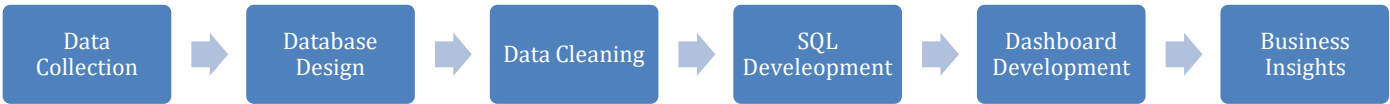
MySQL Workbench	Database design and SQL development	Database creation, ER Diagram (EER), SQL scripting, Views, Stored Procedures, Functions, Indexes, Query Optimization
Microsoft Excel	Interactive dashboard creation and reporting	Pivot Tables, Pivot Charts, Slicers, KPI Cards, Conditional Formatting
Power BI Desktop	Business intelligence dashboard development	Data Modeling, DAX Measures, Interactive Visualizations, Drill-through, Filters, KPI Cards
Tableau Desktop	Visual analytics and dashboard design	Calculated Fields, Interactive Dashboards, Filters, Parameters, Charts and Maps
CSV Files	Source data	Patient, Doctor, Visit, Treatment, Lab Test, Billing, and Insurance datasets imported into MySQL
Microsoft Word	Project documentation	Technical report preparation and project documentation

Project Methodology

The project followed a structured end-to-end analytics workflow beginning with data preparation and ending with business intelligence reporting. The raw healthcare dataset was imported into MySQL, where it was transformed into a normalized relational database. Data integrity was maintained using primary keys, foreign keys, constraints, and normalization techniques.

Analytical SQL queries were then developed to generate operational, clinical, financial, and insurance insights. The validated database served as the central data source for creating interactive dashboards in Microsoft Excel, Power BI, and Tableau. Finally, insights obtained from these dashboards were used to evaluate hospital performance and support data-driven decision-making.

End-to-End Project Workflow



SQL Suite

The SQL Suite forms the core of the Healthcare Analytics System by providing a structured framework for database management, data validation, analytical querying, and performance optimization. MySQL was used to design and implement a normalized relational database capable of efficiently storing and analyzing healthcare data. The SQL implementation includes database creation, table design, data import, validation, exploratory data analysis, advanced SQL techniques, views, stored procedures, functions, and query optimization. These SQL components served as the primary data source for developing interactive dashboards in Microsoft Excel, Power BI, and Tableau.

Database Creation

The Healthcare Analytics database was created using MySQL Workbench to store and manage healthcare information in a centralized relational database. The database was designed to integrate patient records, doctor information, hospital visits, treatments, laboratory tests, billing transactions, insurance policies, insurance providers, and insurance claims into a single analytical environment.

The database was initialized using the following SQL commands:

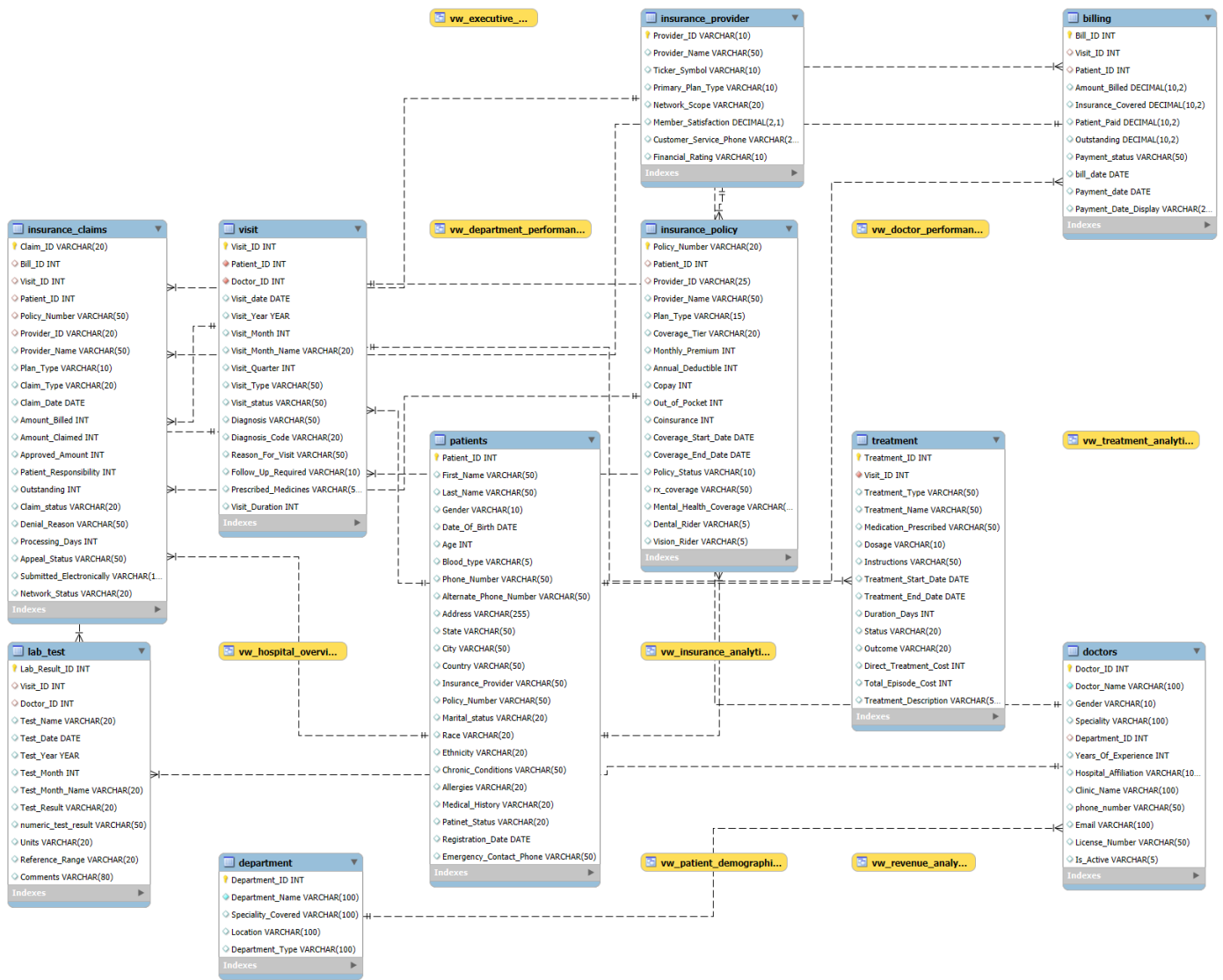
```
# Creating Database
create database HealthcareDB;
use HealthcareDB;
```

Database Design

The Healthcare Analytics database consists of ten normalized tables connected through Primary Keys (PK) and Foreign Keys (FK). The database design follows the principles of Third Normal Form (3NF), reducing redundancy while maintaining data consistency and referential integrity. Each table represents a specific healthcare entity, enabling efficient storage and retrieval of operational, clinical, financial, and insurance information.

Feature	Description
Database Name	HealthcareDB
Database Platform	MySQL Workbench
Number of Tables	10
Normalization	Third Normal Form (3NF)
Primary Keys	Implemented in all tables
Foreign Keys	Used to establish table relationships
Referential Integrity	Maintained using foreign key constraints

Entity Relationship (EER) Diagram of the Healthcare Analytics Database



The Entity Relationship (EER) Diagram represents the logical structure of the **HealthcareDB** database created in MySQL Workbench. It illustrates the relationships between the ten database tables using Primary Keys (PK) and Foreign Keys (FK), ensuring referential integrity and efficient data retrieval.

SQL Development Process

To improve maintainability and readability, the SQL implementation was divided into seventeen separate SQL scripts. Each script performs a specific task within the database development lifecycle, making the project modular and easier to understand.

SQL Project Structure

SQL File	Purpose
01_Database_Setup.sql	Database creation
02_Data_Model.sql	Table creation and relationships
03_Data_Import.sql	Importing CSV datasets
04_Data_Cleaning.sql	Data cleaning and validation
05_Patient_Analysis.sql	Patient analytics
06_Doctor_Analysis.sql	Doctor analytics
07_Department_Analysis.sql	Department analytics
08_Visit_Analysis.sql	Visit analytics
09_Billing_Analysis.sql	Revenue and billing analytics
10_Treatment_Analysis.sql	Treatment analytics
11_Laboratory_Analysis.sql	Laboratory analytics
12_Insurance_Analysis.sql	Insurance analytics
13_Advanced_SQL.sql	CTEs, CASE statements, subqueries
14_Window_Functions.sql	Ranking and cumulative analysis
15_Query_Optimization.sql	Index creation and optimization
16_Views.sql	SQL Views
17_Views_and_Stored_Procedures.sql	Views, Stored Procedures and Functions

Data Cleaning & Validation

Before analysis, the imported healthcare data was thoroughly validated to ensure accuracy, consistency, and completeness. Data cleaning involved verifying data types, checking for duplicate records, validating primary and foreign keys, handling missing values, and ensuring referential integrity across all related tables. These validation procedures ensured that the database was reliable for analytical reporting and dashboard development.

Data Validation Activities

Validation Technique	Purpose
Data Type Validation	Ensured appropriate data types for each attribute
Duplicate Record Check	Prevented duplicate primary key values
NULL Value Validation	Identified and handled missing values
Primary Key Validation	Ensured uniqueness of records
Foreign Key Validation	Maintained referential integrity
Date Validation	Verified correct date formats
Constraint Validation	Enforced business rules within the database

Here are some SQL Queries used in Data Cleaning & Validation:

```
-- handling Duplicates

# 1. Check for duplicate departments
select Department_ID, count(*) as 'Duplicate_Count'
from Department
group by Department_ID
having count(*) > 1;

-- Handling NULL Values

# 11. Find patients with NULL first name
select *
from Patients
where First_Name is null;

-- Data Validation

# 15. Check distinct gender values in Patient
select distinct Gender
from Patients;
```

Exploratory Data Analysis (EDA)

A comprehensive set of SQL queries was developed to analyze operational, clinical, financial, and insurance data. Exploratory Data Analysis (EDA) helped identify trends, evaluate hospital performance, monitor resource utilization, and generate key business metrics. More than one hundred SQL queries were written using joins, aggregate functions, grouping operations, subqueries, and conditional statements.

SQL Analysis Modules

Analysis Module	Description
Patient Analysis	Patient demographics and registration trends
Doctor Analysis	Doctor performance and workload
Department Analysis	Department-wise operational performance
Visit Analysis	Hospital visit trends and patient flow
Billing Analysis	Revenue, collections, and outstanding balances
Treatment Analysis	Treatment frequency and outcomes
Laboratory Analysis	Laboratory test trends and performance
Insurance Analysis	Insurance claims, approvals, and provider performance

Here are some SQL Queries used in EDA:

```
-- Basic Statistics
# 1. Total patients
select count(*) as 'Total_Patients'
from Patients;

# 2. Total doctors
select count(*) as 'Total_Doctors'
from Doctors;

-- Doctor & Department Analysis
# 11. Doctors by department
select
    d.Department_Name,
    count(doc.Doctor_ID) as 'Doctors'
from Department as d
left join Doctors as doc
on d.Department_ID = doc.Department_ID
group by d.Department_Name;

-- Visit Analysis
# 12. Visit type distribution
select Visit_Type, count(*) as 'Visits'
from Visit
group by Visit_Type;

-- Treatment Analysis
# 15. Average treatment cost
select round(avg(Direct_Treatment_Cost), 2) as 'Avg_Treatment_Cost'
from Treatment;

-- Billing Analysis
# 16. Highest bill amount
select max(Amount_Billed) as 'Highest_Bill'
from Billing;

# 17. Payment_Status Distribution
select payment_status, count(*) as 'Bills'
from billing group by payment_status;

-- Insurance Analysis
# 18. Insurance claim status
select Claim_Status, count(*) as 'Claims'
from Insurance_Claims
group by Claim_Status;
```

KPI Design

Several Key Performance Indicators (KPIs) were developed using SQL to measure hospital performance across multiple functional areas. These KPIs served as the foundation for the dashboards created in Microsoft Excel, Power BI, and Tableau.

Operational KPIs	Financial KPIs
Total Patients	Total Revenue
Total Doctors	Amount Collected

Total Departments	Outstanding Amount
Total Visits	Collection Percentage
Total Treatments	Average Bill Amount
Total Laboratory Tests	Insurance Claim Amount
Doctor Workload	Revenue by Department

Advanced SQL Implementation

To enhance analytical capabilities and improve database performance, several advanced SQL concepts were implemented throughout the project. These techniques enabled efficient querying, reusable database objects, and optimized execution of analytical reports.

SQL Feature	Purpose
Joins	Combined data from multiple related tables
Aggregate Functions	Calculated KPIs and summary statistics
GROUP BY & HAVING	Generated summarized reports
CASE Statements	Categorized healthcare records
Subqueries	Performed nested analytical calculations
Common Table Expressions (CTEs)	Improved readability of complex queries
Window Functions	Implemented ranking and running totals
Views	Simplified complex reporting queries
Stored Procedures	Automated repetitive operations
User Defined Functions	Created reusable business logic
Indexes	Improved query execution performance
EXPLAIN	Analyzed query execution plans

Here are SQL Queries Used in Advanced Analysis:

```
-- CTE
```

```
# 1. Top 5 patients by number of visits
```

```
with PatientVisits as (  
  select  
    Patient_ID,  
    count(*) as 'Total_Visits'  
  from Visit  
  group by Patient_ID  
)
```

```
select *  
from PatientVisits  
order by Total_Visits desc  
limit 5;
```

```
-- CTE with Joins
```

```
# 2. Revenue by department
```

```
with DeptRevenue as (  
  select  
    d.Department_Name,  
    sum(b.Amount_Billed) as 'Revenue'  
  from Department as d  
  join Doctors as doc  
  on d.Department_ID = doc.Department_ID  
  join Visit as v  
  on doc.Doctor_ID = v.Doctor_ID  
  join Billing as b  
  on v.Visit_ID = b.Visit_ID  
  group by d.Department_Name  
)
```

```
select *  
from DeptRevenue  
order by Revenue desc;
```

```
-- Subqueries
# 9. Department with highest revenue
select
Department_Name,
Revenue
from
(select
d.Department_Name,
sum(b.Amount_Billed) as 'Revenue'
from Department as d
join Doctors as doc
on d.Department_ID = doc.Department_ID
join Visit as v
on doc.Doctor_ID = v.Doctor_ID
join Billing as b
on v.Visit_ID = b.Visit_ID
group by d.Department_Name) as x
where
Revenue = (select max(Revenue)
from
(select
sum(b.Amount_Billed) as 'Revenue'
from Department as d
join Doctors as doc
on d.Department_ID = doc.Department_ID
join Visit as v
on doc.Doctor_ID = v.Doctor_ID
join Billing as b
on v.Visit_ID = b.Visit_ID
group by d.Department_Name) as y);
```

```
-- Window Functions
```

```
# 1. Assign row number to patients by registration date
```

```
select
Patient_ID,
concat(First_Name, ' ', Last_Name) as 'Patient_Name',
Registration_Date,
row_number() over(order by Registration_Date) as 'Row_Num'
from Patients;
```

```
# 2. Rank doctors by revenue
```

```
select
d.Doctor_Name,
sum(b.Amount_Billed) as 'Revenue',
rank() over(order by sum(b.Amount_Billed) desc) as 'Revenue_Rank'
from Doctors as d
join Visit as v
on d.Doctor_ID = v.Doctor_ID
join Billing as b
on v.Visit_ID = b.Visit_ID
group by d.Doctor_Name;
```

-- Dashboard Views

1. Hospital overview

```
create view vw_hospital_overview as
select
v.Visit_ID,
v.Visit_Date,
v.Visit_Type,
v.Visit_Status,
v.Diagnosis,
p.Patient_ID,
concat(p.First_Name, ' ', p.Last_Name) as 'Patient_Name',
timestampdiff(year, p.Date_of_Birth, curdate()) as 'Age',
p.Gender,
d.Doctor_Name,
dep.Department_Name,
b.Amount_Billed,
b.Patient_Paid,
b.Insurance_Covered,
b.Outstanding
from Visit as v
join Patients as p
on v.Patient_ID = p.Patient_ID
join Doctors as d
on v.Doctor_ID = d.Doctor_ID
join Department as dep
on d.Department_ID = dep.Department_ID
left join Billing as b
on v.Visit_ID = b.Visit_ID;

select *
from vw_hospital_overview;
```

Query Optimization

Database performance was enhanced through query optimization techniques. Indexes were created on frequently searched columns to reduce query execution time and improve retrieval efficiency. The EXPLAIN statement was used to analyze execution plans and verify that indexes were utilized effectively during query processing.

Optimization Techniques

- Created indexes on frequently queried columns.
- Improved query execution using EXPLAIN.
- Optimized JOIN operations.
- Implemented Views for reusable reporting.
- Used Stored Procedures to automate repetitive tasks.

Here are some codes used in Optimization:

```
-- Query Optimization & Indexing

# 1. Analyze execution plan for patient lookup
explain
select *
from Patients
where Patient_ID = 101;

# 2. Create index on Visit.Patient_ID
create index idx_visit_patient
on Visit(Patient_ID);

# 3. Create index on Visit.Doctor_ID
create index idx_visit_doctor
on Visit(Doctor_ID);

# 4. Create index on Billing.Bill_Date
create index idx_billing_bill_date
on Billing(Bill_Date);

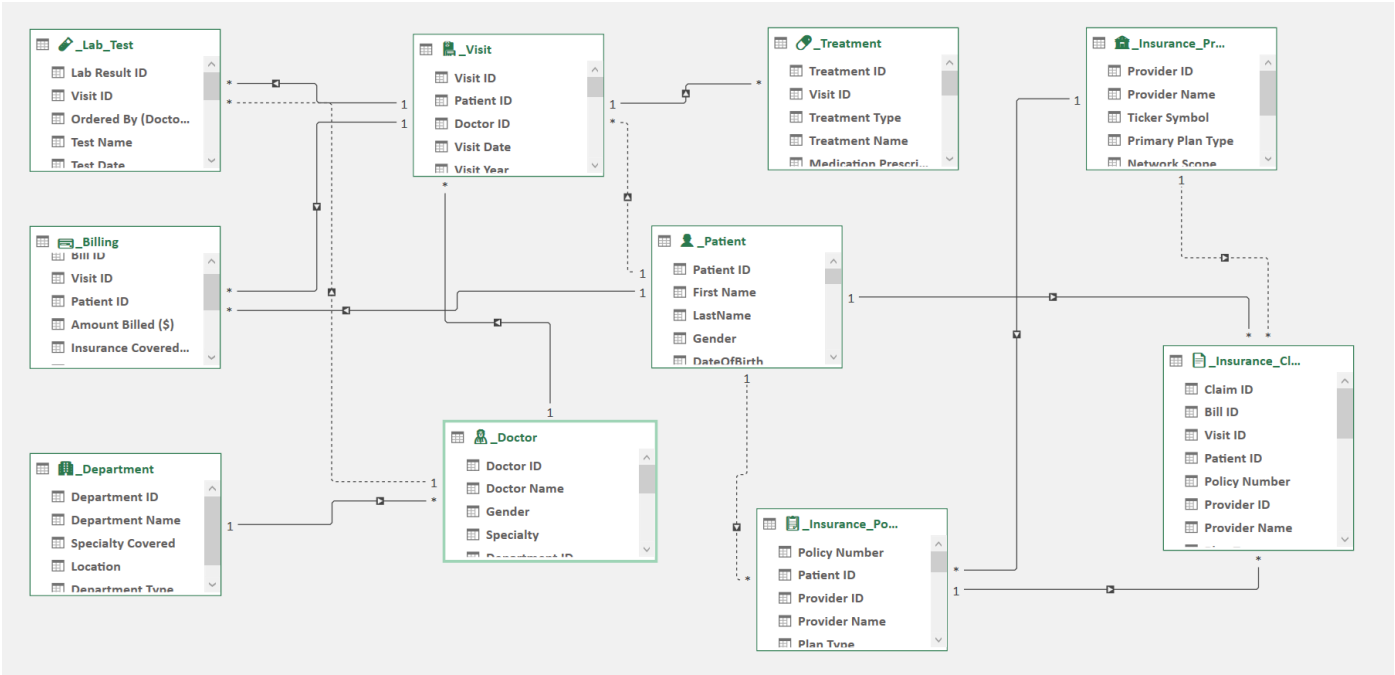
# 5. Verify optimized query using index
explain
select *
from Visit
where Patient_ID = 101;
```

Summary

The SQL Suite serves as the backbone of the Healthcare Analytics System by integrating database design, data validation, analytical querying, and performance optimization into a unified framework. The implementation of relational database principles, advanced SQL techniques, and optimization strategies enabled efficient analysis of healthcare data while providing a reliable foundation for dashboard development in Microsoft Excel, Power BI, and Tableau. The SQL implementation successfully transformed raw healthcare records into meaningful business insights that support informed decision-making and operational excellence.

Excel Suite

Excel Data Model Overview



The Excel dashboards are powered by a relational data model consisting of ten interconnected tables imported into the Excel Data Model. Relationships were established using unique identifiers such as **Patient ID**, **Doctor ID**, **Visit ID**, **Department ID**, **Provider ID**, and **Policy Number**, enabling seamless integration across operational, clinical, billing, and insurance datasets.

The centralized data model ensures consistency in calculations, eliminates duplicate data, and supports cross-table analysis through Pivot Tables and Pivot Charts.

Excel Dashboard Development

Microsoft Excel was used to design interactive dashboards for analyzing hospital operations, patient care, and financial performance. The dashboards were developed using Pivot Tables, Pivot Charts, Slicers, Timelines, Conditional Formatting, and KPI Cards, enabling users to explore healthcare data through dynamic visualizations.

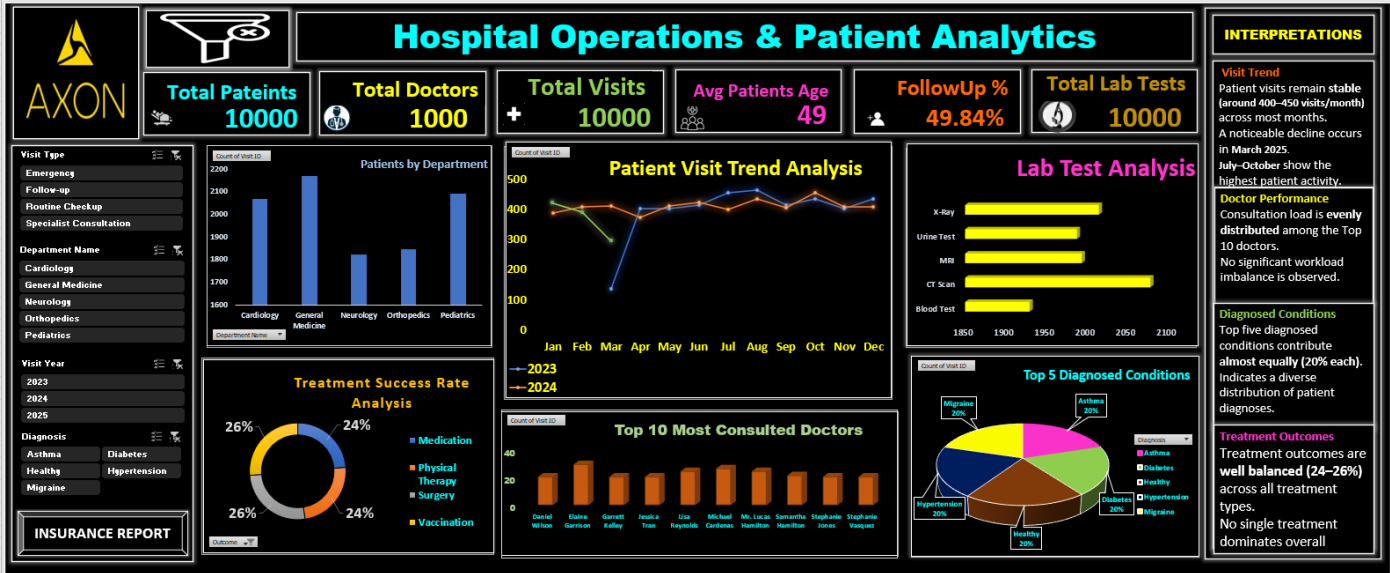
The dashboards provide hospital administrators with a consolidated view of operational, clinical, and financial metrics, allowing efficient monitoring of key performance indicators and supporting informed decision-making.

Dashboard Architecture

The Excel reporting solution consists of two interactive dashboards:

Dashboard	Purpose
Hospital Operations & Patient Analytics	Monitors patient demographics, hospital visits, laboratory tests, doctor performance, treatment outcomes, and diagnosed conditions.
Financial & Insurance Analytics	Analyzes hospital revenue, billing performance, insurance policies, claims, provider performance, and payment analysis.

Dashboard 1 - Hospital Operations & Patient Analytics



Dashboard Overview

The Hospital Operations & Patient Analytics dashboard provides a comprehensive overview of patient activity, doctor performance, laboratory testing, and treatment outcomes across the healthcare organization. Interactive slicers allow users to filter information based on visit type, department, diagnosis, and year, enabling detailed operational analysis.

Key Performance Indicators

KPI	Description
Total Patients	Total number of registered patients
Total Doctors	Number of doctors available
Total Visits	Overall patient visits
Average Patient Age	Average age of patients
Follow-up Percentage	Percentage of follow-up visits
Total Lab Tests	Total laboratory tests performed

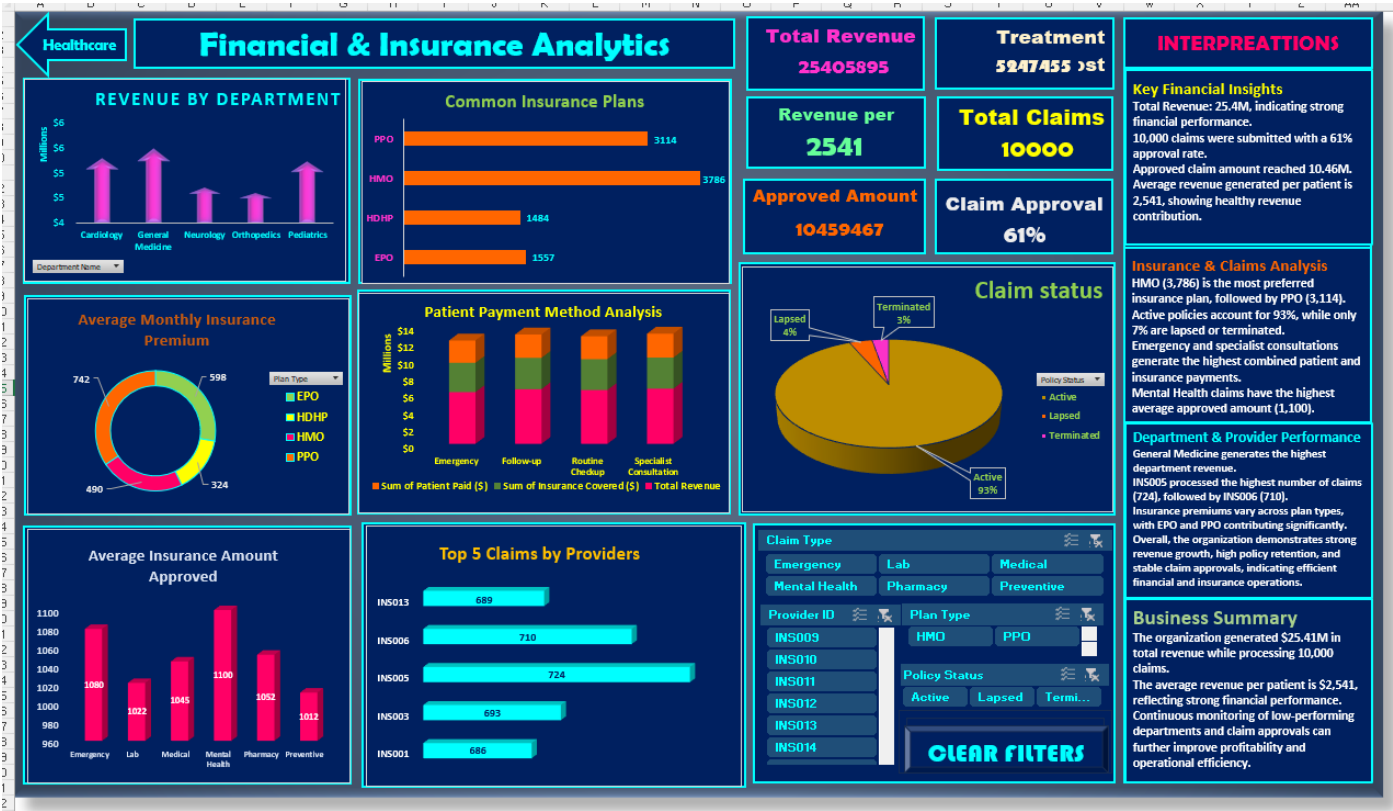
Dashboard Visualizations

Visualization	Business Purpose
Patients by Department	Compare patient distribution across departments
Patient Visit Trend	Analyze monthly patient visits over different years
Lab Test Analysis	Identify frequently performed laboratory tests
Treatment Success Rate	Compare treatment outcomes
Top 10 Most Consulted Doctors	Identify doctors with highest consultation volume
Top Diagnosed Conditions	Determine the most common medical conditions

Business Insights

- Patient visits remain consistent throughout the year.
- General Medicine records the highest patient volume.
- CT Scan is the most frequently performed laboratory test.
- Treatment outcomes are evenly distributed across treatment categories.
- Doctor workload is balanced across departments.

Dashboard 2 – Financial & Insurance Analytics



Dashboard Overview

The Financial & Insurance Analytics dashboard focuses on hospital revenue, insurance coverage, claim processing, provider performance, and patient payment analysis. The dashboard enables financial teams to monitor revenue generation, insurance efficiency, and claim approval performance.

Key Performance Indicators (KPIs)

KPI	Description
Total Revenue	Overall hospital revenue generated
Revenue per Patient	Average revenue generated per patient
Treatment Revenue	Revenue generated through treatments
Total Claims	Number of insurance claims submitted
Approved Amount	Total claim amount approved
Claim Approval Rate	Percentage of approved insurance claims

Dashboard Visualizations

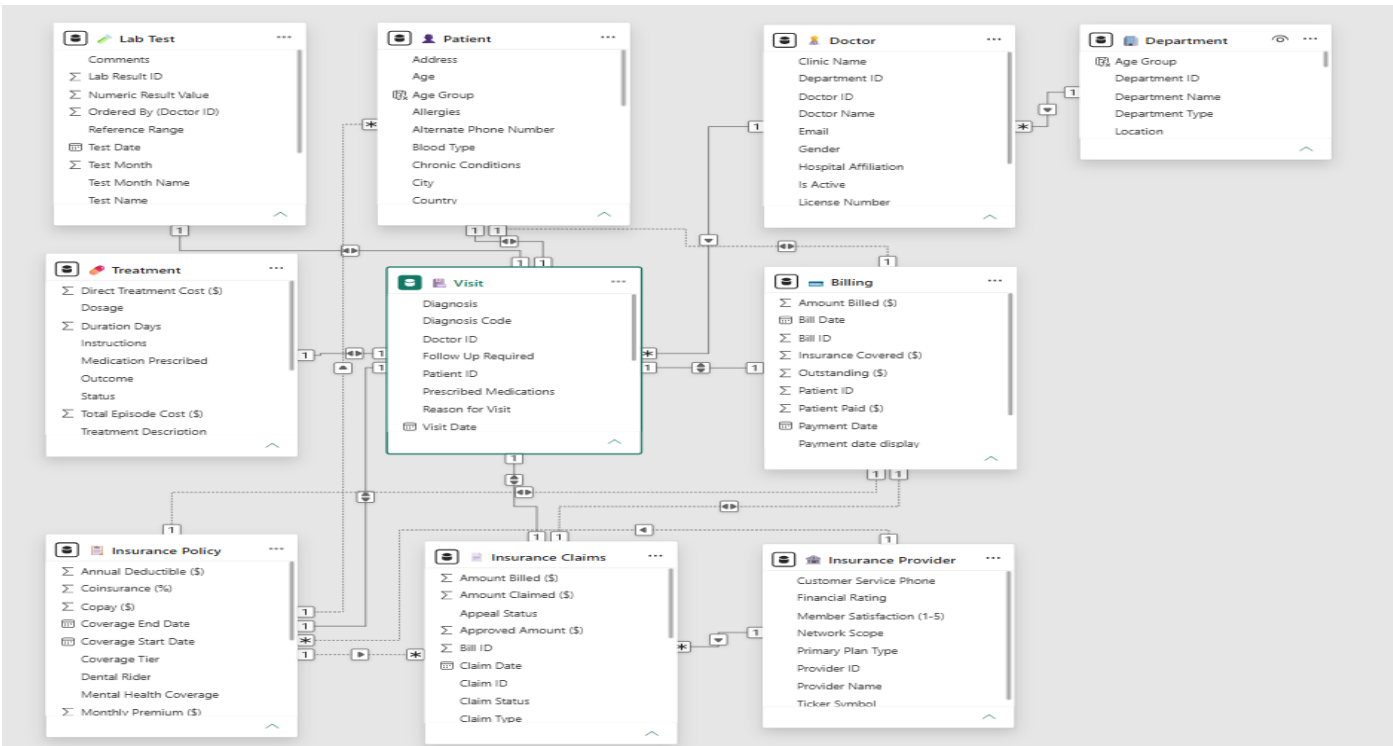
Visualization	Business Purpose
Revenue by Department	Compare department-wise revenue
Common Insurance Plans	Analyze popular insurance plans
Monthly Insurance Premium	Compare premium amounts
Patient Payment Analysis	Compare insurance-covered and patient-paid amounts
Claim Status	Monitor active, lapsed, and terminated policies
Average Insurance Amount Approved	Compare approved amounts by claim type
Top 5 Claims by Providers	Identify providers with the highest number of claims

Business Insights

- General Medicine generates the highest revenue.
- Active insurance policies account for the majority of claims.
- HMO is the most frequently selected insurance plan.
- Insurance approval rate exceeds 60%.
- Provider INS005 processed the highest number of claims.

Power BI Suite

Power BI Data Model Overview



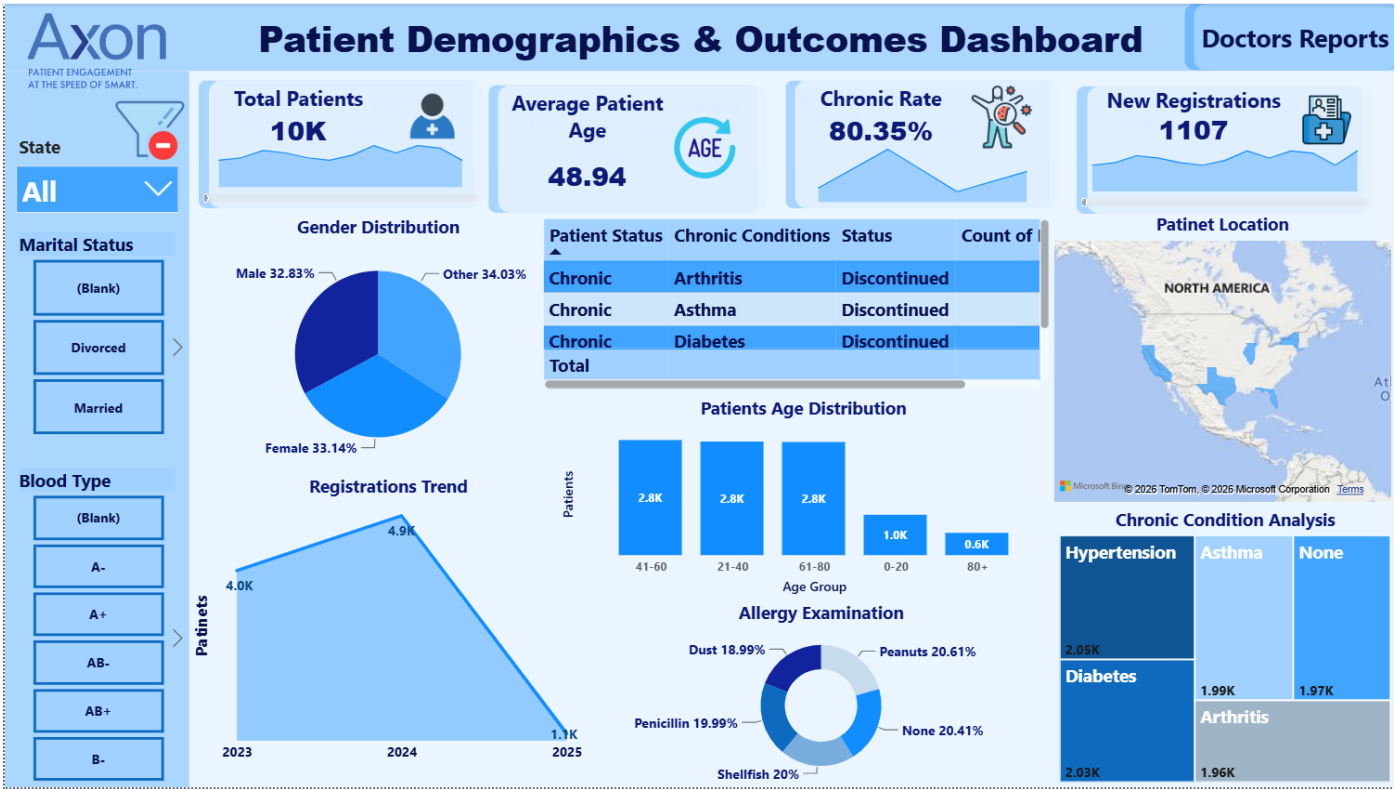
The Power BI data model was developed using a star-schema inspired relational structure to ensure efficient data analysis and visualization. The model consists of ten interconnected tables imported from the MySQL database. Relationships

between tables were established using primary and foreign keys, enabling seamless cross-table filtering and accurate KPI calculations.

The data model acts as the semantic layer for the dashboards by organizing patient, doctor, visit, treatment, billing, laboratory, and insurance data into a unified analytical model.

Power BI Dashboard Overview

Dashboard 1 : Patient Demographics & Outcomes Dashboard



Purpose

This dashboard provides an overview of patient demographics, registration trends, chronic disease prevalence, allergy distribution, and geographical patient distribution. Interactive slicers enable users to analyze data by state, marital status, and blood type.

KPIs

KPI	Description
Total Patients	Total number of registered patients
Average Patient Age	Average age of all patients
Chronic Disease Rate	Percentage of patients with chronic conditions
New Registrations	Newly registered patients

Visualizations Included

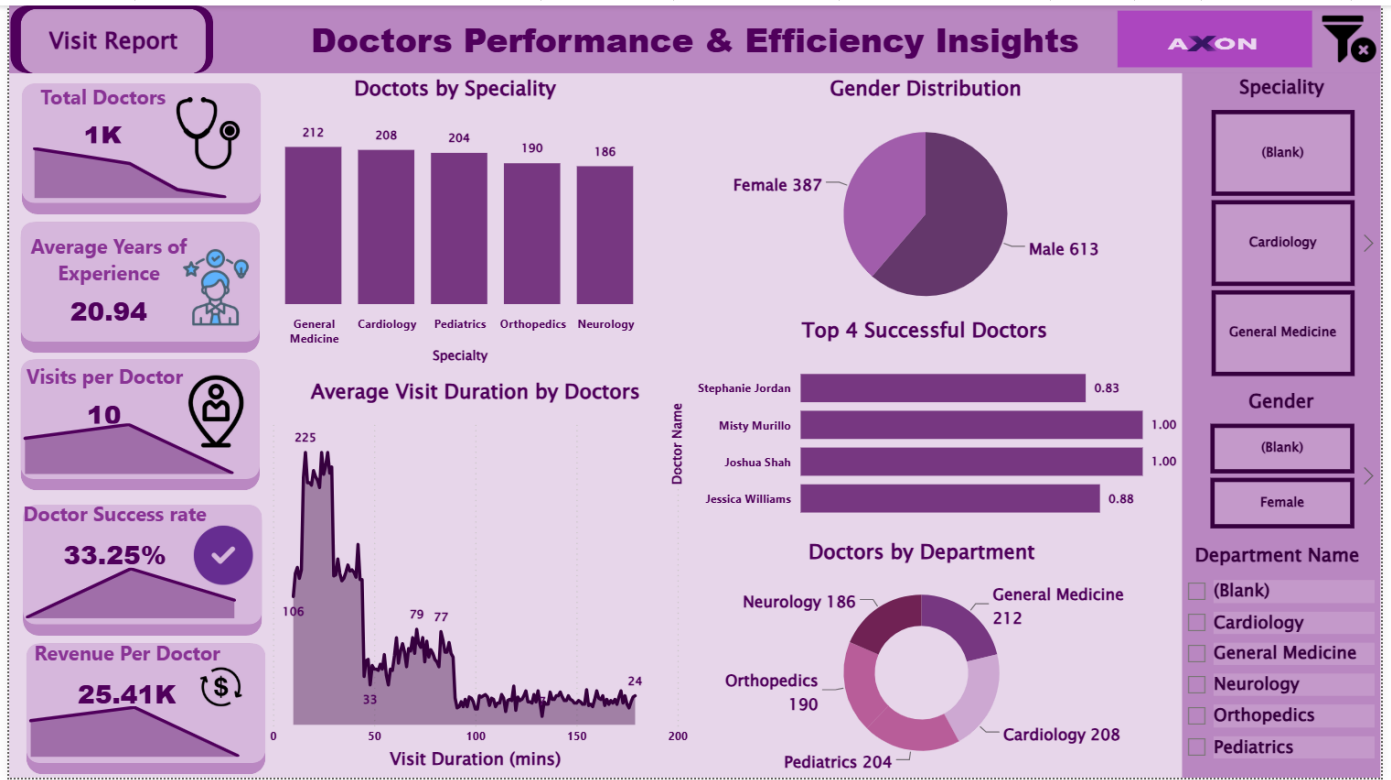
Visualization	Purpose
Gender Distribution (Pie Chart)	Analyze patient gender composition
Patient Status Table	View patient conditions and treatment status

Patient Location Map	Geographic distribution of patients
Registration Trend (Area Chart)	Monitor yearly patient registrations
Age Distribution (Column Chart)	Analyze patients by age groups
Allergy Examination (Donut Chart)	Distribution of allergy types
Chronic Condition Analysis (Treemap)	Compare major chronic diseases

Key Insights

- The hospital recorded **10,000 patient visits** across five departments with a balanced distribution of patient volume.
- General Medicine and Pediatrics handled the highest number of patients, indicating greater service demand.
- Laboratory testing and treatment outcomes remained consistent, reflecting stable clinical operations.
- Follow-up visits accounted for nearly **50%** of total visits, highlighting effective patient monitoring.

Dashboard 2: Doctors Performance & Efficiency Dashboard



Purpose

This dashboard evaluates doctor productivity and departmental performance by analyzing doctor workload, consultation duration, specialization, revenue contribution, and success rate.

KPIs Displayed

KPI	Description
Total Doctors	Total doctors available
Average Years of Experience	Average professional experience

Visits per Doctor	Average patient visits handled
Doctor Success Rate	Treatment success percentage
Revenue per Doctor	Average revenue generated by each doctor

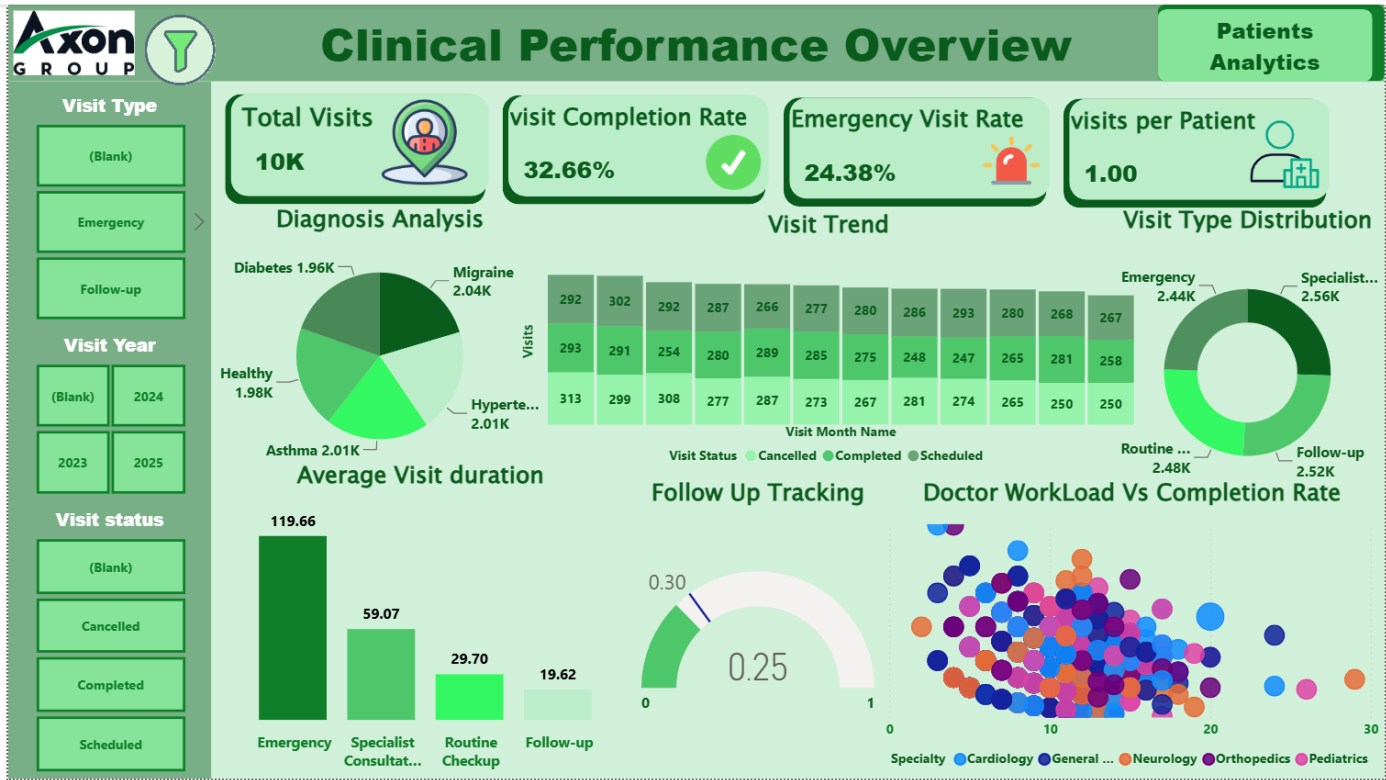
Visualizations Included

Visualization	Purpose
Gender Distribution (Pie Chart)	Analyze patient gender composition
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Age Distribution (Column Chart)	Analyze patients by age groups
Allergy Examination (Donut Chart)	Distribution of allergy types
Chronic Condition Analysis (Treemap)	Compare major chronic diseases

Key Insights

- The hospital generated **strong overall revenue** with a healthy insurance claim approval rate.
- Most patients were covered under **HMO and PPO** insurance plans, contributing significantly to revenue.
- General Medicine produced the highest departmental revenue and insurance claims.
- The low percentage of terminated and lapsed policies indicates effective insurance management.

Dashboard 3: Clinical Performance Overview



Purpose

This dashboard focuses on hospital operational efficiency by analyzing patient visits, diagnosis patterns, treatment completion, follow-up tracking, and doctor workload.

KPIs

KPI	Description
Total Visits	Total patient visits
Visit Completion Rate	Percentage of completed visits
Emergency Visit Rate	Percentage of emergency visits
Visits per Patient	Average visits per patient

Visualizations Included

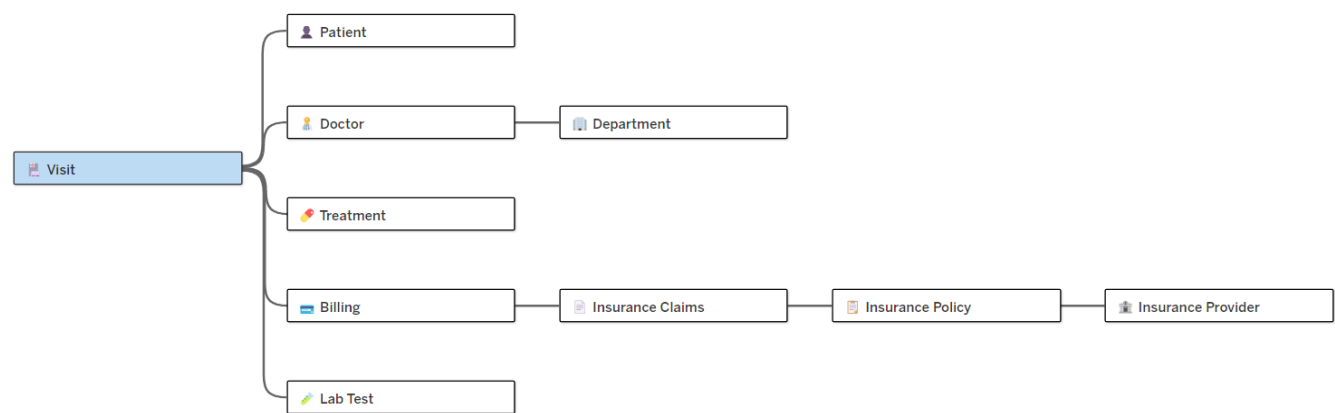
Visualization	Purpose
Diagnosis Analysis (Pie Chart)	Distribution of diagnosed conditions
Visit Trend (Heatmap)	Monthly visit performance
Visit Type Distribution (Donut Chart)	Breakdown of visit categories
Average Visit Duration (Column Chart)	Compare duration by visit type
Follow-up Tracking (Gauge Chart)	Monitor follow-up completion
Doctor Workload vs Completion Rate (Scatter Plot)	Analyze doctor performance and workload

Key Insights

- The workload was distributed evenly among doctors across all specialties.
- General Medicine had the highest number of doctors, supporting higher patient demand.
- Successful doctors demonstrated high treatment completion rates with consistent patient outcomes.
- Average doctor experience and visit duration indicate efficient utilization of medical resources.

Tableau Suite

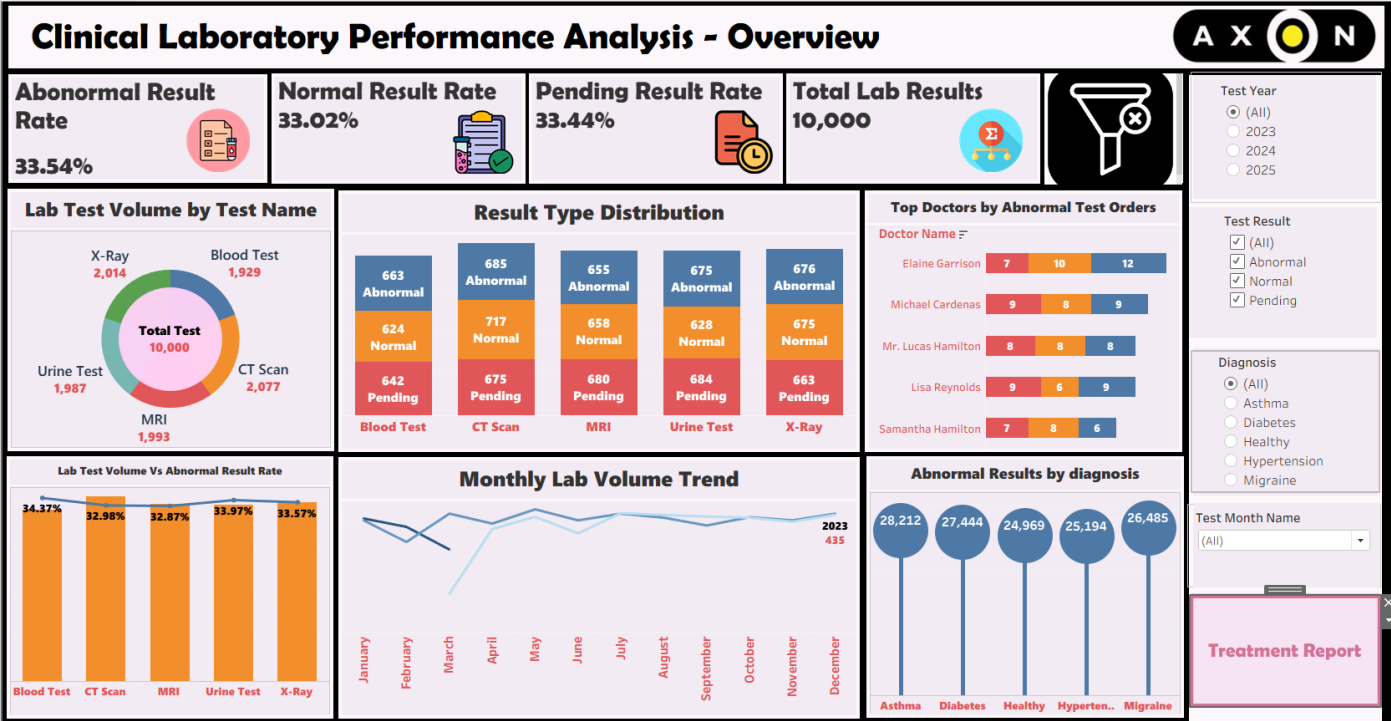
Tableau Data Model



Description

The Tableau dashboards are built using a relational data model imported from the Healthcare Analytics database. The model consists of ten interconnected tables linked through primary and foreign keys, enabling seamless analysis of patient demographics, treatments, laboratory performance, billing, insurance, and hospital operations. The centralized model ensures data consistency across all Tableau dashboards while supporting interactive filtering and drill-down analysis.

Dashboard 1 – Clinical Laboratory Performance Analysis



Dashboard Overview

This dashboard monitors laboratory operations by tracking test volumes, result status, abnormal findings, monthly trends, and doctor-wise laboratory requests. It enables hospital administrators to evaluate laboratory efficiency and diagnostic performance.

KPIs

KPI	Description
Abnormal Result Rate	Percentage of abnormal laboratory results
Normal Result Rate	Percentage of normal laboratory results
Pending Result Rate	Percentage of pending reports
Total Lab Results	Total laboratory tests performed

Visualizations Included

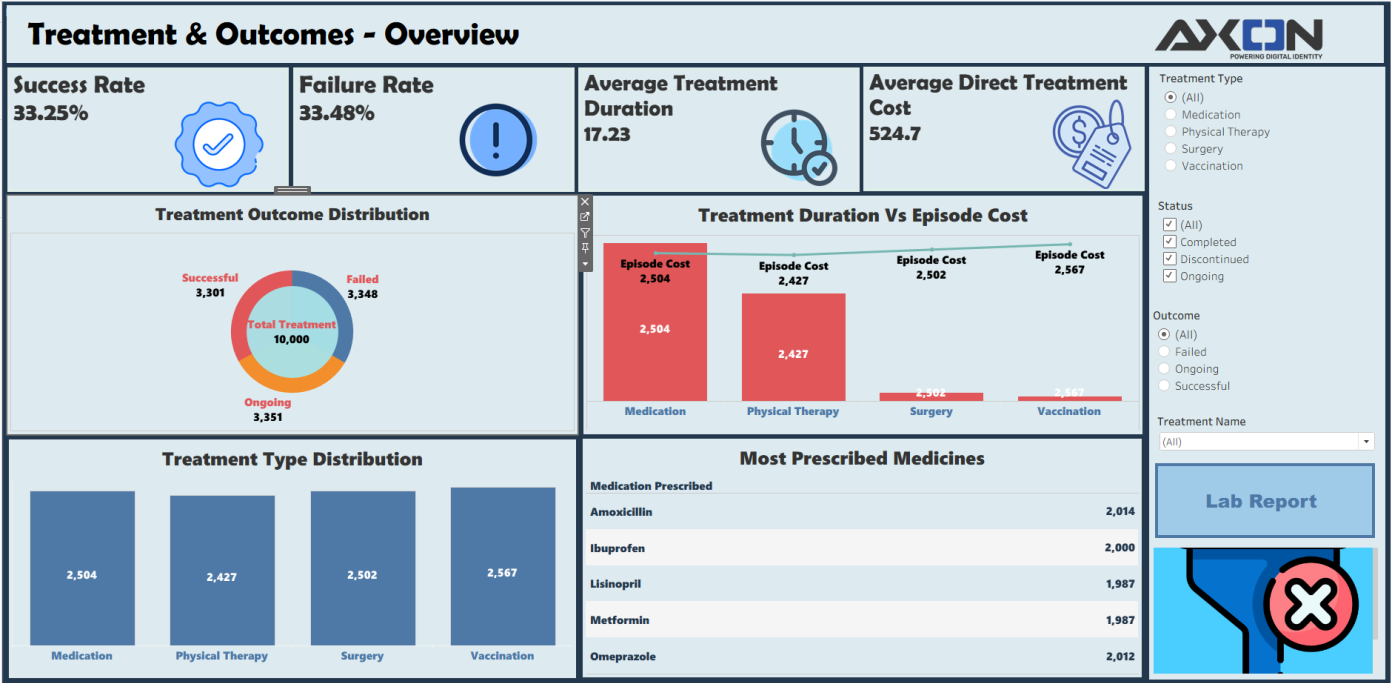
Visualization	Purpose
Lab Test Volume by Test Name (Donut Chart)	Compare test volumes
Result Type Distribution (Stacked Column Chart)	Analyze result status by test

Top Doctors by Abnormal Test Orders (Bar Chart)	Identify doctors ordering abnormal tests
Lab Test Volume vs Abnormal Rate	Compare abnormal percentages across tests
Monthly Lab Volume Trend (Line Chart)	Track laboratory workload over time
Abnormal Results by Diagnosis (Bubble Chart)	Compare abnormal findings by diagnosis
Interactive Filters	Filter by year, diagnosis, test result, and month

Key Insights

- A total of **10,000 laboratory tests** were performed during the analysis period.
- CT Scan and X-Ray recorded the highest testing volumes.
- Abnormal, normal, and pending results are distributed almost equally.
- Laboratory workload remained consistent throughout the year.

Dashboard 2 - Treatment & Outcomes Overview



Average Treatment Duration	Average duration of treatments
Average Direct Treatment Cost	Average treatment expenditure

Visualizations Included

Visualization	Purpose
Treatment Outcome Distribution (Donut Chart)	Compare successful, failed, and ongoing treatments
Treatment Duration vs Episode Cost	Analyze treatment cost across treatment types
Treatment Type Distribution (Bar Chart)	Compare treatment categories
Most Prescribed Medicines (Table)	Identify frequently prescribed medications
Interactive Filters	Filter by treatment type, outcome, and status

Key Insights

- Treatment outcomes are evenly distributed between successful, failed, and ongoing treatments.
- Medication is the most frequently performed treatment type.
- Average treatment duration remains stable across treatment categories.
- Frequently prescribed medications indicate standardized clinical practices.

Business Insights

The Healthcare Analytics System provides comprehensive insights into hospital operations by integrating patient, clinical, financial, laboratory, and insurance data into interactive dashboards. The analysis enables healthcare organizations to monitor performance, identify trends, and support data-driven decision-making.

Key Business Insights

- The hospital managed **10,000 patient visits** across multiple departments, demonstrating consistent healthcare service utilization.
- **General Medicine** recorded the highest patient volume, indicating greater demand for primary healthcare services.
- Patient registrations remained relatively stable throughout the analysis period, reflecting consistent patient inflow.
- Treatment outcomes were evenly distributed among successful, ongoing, and failed treatments, indicating balanced clinical performance.
- Laboratory testing volumes remained consistent, with CT Scan and X-Ray among the most frequently performed diagnostic tests.
- Insurance claim approval rates were high, demonstrating efficient insurance claim processing and financial management.
- Revenue generation remained consistent across departments, contributing to stable financial performance.
- Interactive dashboards enabled real-time monitoring of KPIs, improving operational visibility and supporting informed decision-making.

Recommendations

Based on the analytical findings obtained from the Healthcare Analytics System, the following recommendations are suggested to improve operational efficiency and healthcare service delivery:

- Implement predictive analytics to identify patients at high risk of chronic diseases and hospital readmissions.
- Introduce automated patient follow-up reminders to improve treatment adherence and appointment completion rates.
- Optimize doctor workload by balancing patient distribution across departments.
- Continuously monitor high-cost treatments and laboratory procedures to improve cost efficiency.
- Strengthen preventive healthcare programs to reduce the prevalence of chronic diseases.
- Expand dashboard capabilities by integrating real-time hospital information systems.
- Regularly monitor insurance claim processing to minimize delays and improve claim approval efficiency.
- Develop executive-level dashboards to support strategic planning and organizational performance monitoring.

Challenges & Solutions

During the development of the Healthcare Analytics System, several technical and analytical challenges were encountered. Appropriate solutions were implemented to ensure successful project completion.

Challenge Encountered	Solution Implemented
Managing large healthcare datasets	Designed a normalized relational database (Third Normal Form - 3NF) to organize data efficiently, eliminate redundancy, and improve maintainability.
Ensuring data integrity and consistency	Implemented Primary Keys, Foreign Keys, constraints, and referential integrity to maintain accurate relationships between database tables.
Importing and organizing multiple datasets	Utilized the MySQL Table Data Import Wizard to import CSV files into structured database tables while validating record counts and data types.
Developing complex analytical queries	Applied advanced SQL concepts such as Joins, Aggregate Functions, CASE Statements, Subqueries, Common Table Expressions (CTEs), Window Functions, Views, Stored Procedures, and User-Defined Functions to perform comprehensive data analysis.
Optimizing query performance	Created indexes on frequently accessed columns and analyzed query execution plans using the EXPLAIN statement to improve database performance.
Designing interactive dashboards	Developed dynamic dashboards in Microsoft Excel, Power BI, and Tableau using KPI cards, charts, slicers, filters, and interactive visualizations to enhance user experience.
Maintaining consistency across reporting platforms	Used the validated MySQL database as the single source of truth, ensuring consistent KPIs and business metrics across Excel, Power BI, and Tableau dashboards.

Presenting complex healthcare data effectively	Converted raw healthcare data into intuitive dashboards using charts, maps, tables, and KPI indicators, enabling stakeholders to interpret insights quickly and support data-driven decision-making.
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Conclusion

The Healthcare Analytics System successfully demonstrates an end-to-end business analytics solution by integrating MySQL, Microsoft Excel, Power BI, and Tableau into a unified healthcare reporting platform. The project transformed raw healthcare datasets into a structured relational database and utilized SQL-based analytical techniques to generate meaningful operational, clinical, financial, laboratory, and insurance insights.

Interactive dashboards developed across multiple business intelligence platforms enabled effective visualization of key performance indicators and supported comprehensive analysis of hospital performance. The implementation of relational database design, advanced SQL programming, and modern visualization tools ensured accurate reporting, improved data accessibility, and enhanced decision-making capabilities.

Overall, the project highlights the practical application of data analytics in the healthcare sector and demonstrates how business intelligence technologies can improve operational efficiency, resource utilization, financial management, and patient care. The developed system provides a scalable foundation for future enhancements, including predictive analytics, real-time reporting, and AI-driven healthcare decision support.